

Component fun

This is the time to be for connected appliance makers because almost every industry major is into the home automation game, and they are busy churning out amazing chips and components for the same. Here are some examples:

- For smart home gateway systems, Intel offers PUMA DOCSIS 3.1 family of systems-on-chips (SoCs) for cable and Intel AnyWAN GRX network processor for telecom networks.
- Intel's RealSense technology is a 3D camera system that can track motion and gestures of users more effectively.
- Ambarella has introduced S3L SoC family that brings the low bit-rate benefits of H.265 to both Cloud based home monitoring and mainstream professional cameras.
- Atmel's Smart BTLC1000 is an ultra-low power Bluetooth Smart (BLE 4.1) SoC that features 30 per cent or more power saving in dynamic mode compared to available solutions, taking less than 4mA in RX and less than 3mA in TX at 0dbm.
- Marvell's Andromeda Box IoT platform can be applied to IoT gateways and smarthome devices. The company recently introduced 88W8977, an SoC that includes both 2.4GHz/5GHz wireless local area network and Bluetooth 4.2. It also supports 802.11mc and BLE Angle of Arrival (BLE AoA) based location for indoor navigation.
- MediaTek is also selling IoT chips with low power consumption and high-level integration such as MT7697 chipset that supports BLE and Wi-Fi connectivity for short-range and long-range data transmission, respectively.
- Qualcomm Technologies' Snapdragon 212 Smart Home Reference Platform combines computing, voice recognition, audio, display,

camera, connectivity and control capabilities for smart home control hubs, speakers, home appliances and multimedia devices.

- RealTek Semiconductor's RTL8762A is a BLE 4.2 SoC that can be used in Bluetooth remotes for controlling TVs, set-top boxes, air-conditioners and other smart appliances. It requires no directional pointing of the controller device. Bluetooth Voice adds voice capability to the remote.
- Toshiba's TC35667WBG-006 is a communication integrated chip (IC) that uses Bluetooth 4.1 and is suitable for developing Scatternet devices. Scatternet standard defines two functions in the Bluetooth core specification: devices have to support either multiple or simultaneous connections with master and slave devices, or multiple connections between two or more master devices at the same time. Toshiba's new IC supports both functions with BLE, making it easy to create a network with very small devices. TC35667WBG-006 is configured with read-only memory (ROM) software for Scatternet and multi-point connections. Its functionality and small form factor makes it ideal for low-profile applications like contactless cards and remotes.
- The future is bound to get all the more exciting with 3D-printed circuitry from Xerox. The company has shown that, with its 3D printing technology, it is possible to create 3D circuitry within a 3D-printed object as it is being manufactured, with sensors, actuators, coils and such embedded parts directly into the device. This removes barriers of shape and rigidity, enabling an incredibly diverse range of smartdevices, which can sense and interact with the environment.

of the popular platforms, then it becomes simple to set up and use, too. However, can security be taken for granted as much as smartness and simplicity?

During a recent research on the security of Internet-connected home devices, doctoral students at Princeton University, USA, found that several popular devices were leaking information such as zip codes of users, by sending these over the Internet without proper encryption.

In another similar venture that involved attacking an experimental setup of Samsung SmartThings home automation platform with a lock-pick malware app, cyber security researchers at University of Michigan, USA, were able to get the pass-code to open a home's front door.

A few other experimental attacks levelled against the system also succeeded. The results led a professor to comment in a media report that home automation is okay as a

Which language does your device speak

One of the conundrums facing the smart home space is the lack of one universal protocol. Different devices use different protocols, chosen on the basis of what suits its function best. While this could lead to headaches for the user, till such time a clear winner is known, we have to be familiar with all the popular options. So here is a list of some of the contenders:

- KNX
- X10
- Insteon
- Z-Wave
- ZigBee
- Wi-Fi
- Bluetooth and BLE
- Thread
- NFC
- ULE DECT

hobby but not yet ready for regular use, considering the poor state of security of a popular system that has been in use for several years.

Experts also feel that smart TVs

are a sure-shot route for hackers to reach your personal computers and trace out sensitive information like passwords. In a recent media interview, Pam Dixon, executive director of World Privacy Forum, commented, "As more people get these smartsets, we will see a lot more hacking of TVs to get into the home computer network. It is not a backdoor or a sidedoor, it is a front door to your home network. It is a toboggan run directly to your computer."

In a shocking demonstration, Candid Wueest, a threat researcher with digital security firm Symantec, successfully infected his Android based smart TV with ransomware, a piece of malicious software that locks a computing device until a user pays a ransom to the hacker.

Attempting to overcome such security threats, Samsung has included a security suite called GAIA in the Tizen based OS used in its